RETINA AI
HACKATHON 2026

STUDENT DROPOUT PREDICTION

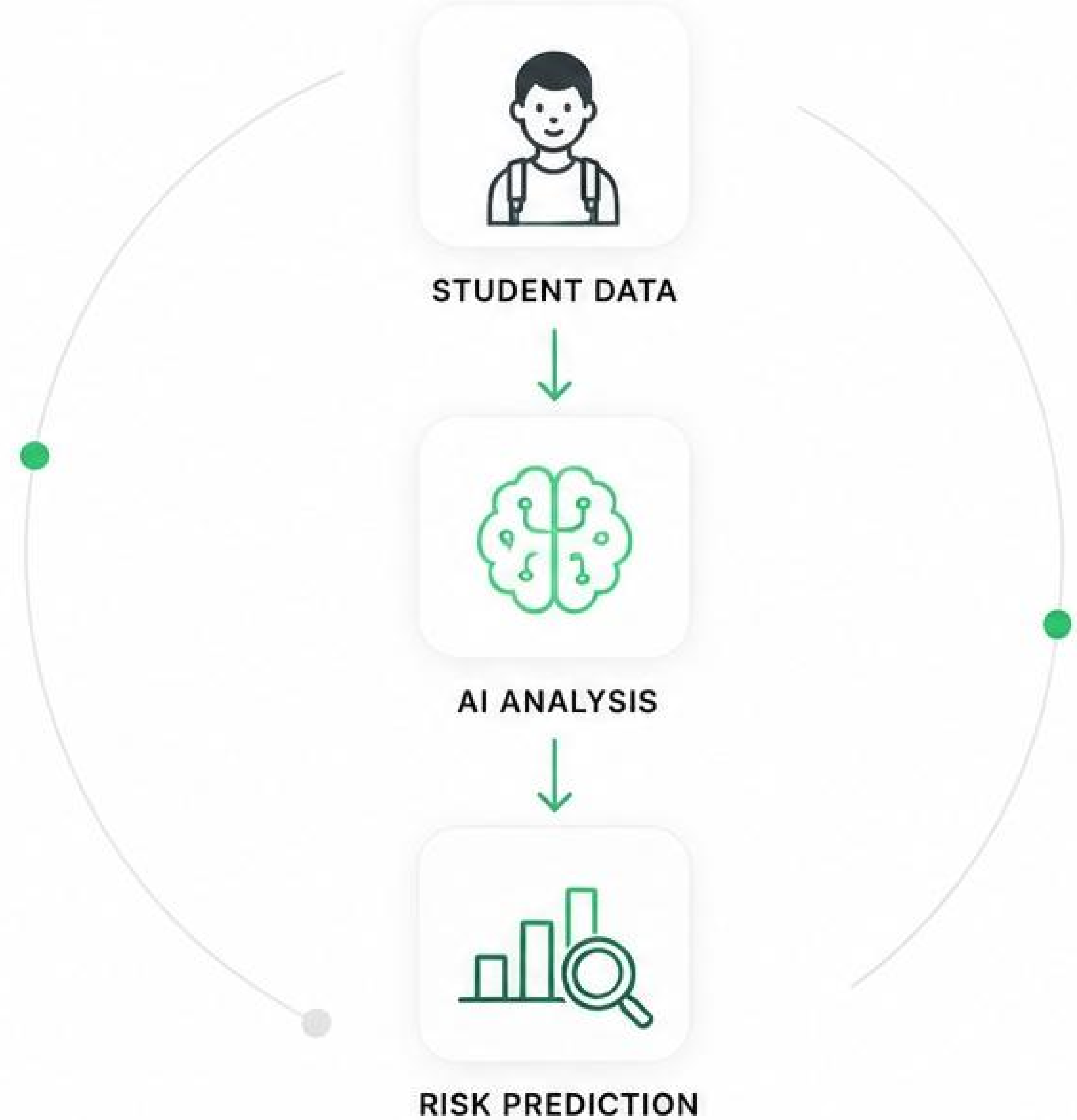
AI-powered early warning system for identifying at-risk students

ATTENDANCE | ACADEMICS | COUNSELLOR NOTES

BEST MODEL PERFORMANCE

0.7214 MACRO F1

+0.0875
OVER BEST SINGLE MODEL



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PROBLEM STATEMENT

Student dropout is a critical challenge affecting academic institutions and student success.

> The goal is to identify at-risk students early so that timely interventions can help them stay on track and succeed.

KEY CHALLENGES

HIGH DROPOUT RATES

Many students drop out due to academic, financial, or personal challenges.

LATE DETECTION

At-risk students are often identified too late for effective intervention.

DISPARATE DATA SOURCES

Data is scattered across attendance, academics, and counsellor notes.

LIMITED INSIGHTS

Institutions lack proactive tools to predict and prevent dropouts.

DATASET OVERVIEW

Multi-modal student data collected from academic records, attendance history, and counsellor interactions.

The attendance dataset alone contained **more than 1 million records**, making feature engineering a key part of the project.

12,000
TRAINING STUDENTS

1,048,575
ATTENDANCE RECORDS

3,000
TEST STUDENTS

15,000
COUNSELLOR NOTES

3 DATA MODALITIES

Academics • Attendance • NLP

DATA PREPROCESSING

Raw data from multiple sources was cleaned, transformed, and standardized to ensure high-quality input for model training.

CLEAN. TRANSFORM. STANDARDIZE.

Ensuring reliable, consistent and model-ready data.

PREPROCESSING PIPELINE

1 DATA CLEANING

- Removed duplicates
- Handled missing values
- Fixed inconsistent entries

2 OUTLIER HANDLING

- Detected outliers using IQR
- Capped extreme values

3 DATA TRANSFORMATION

- Encoding categorical variables
- Aggregated attendance by student
- Date to numerical features

4 FEATURE SCALING

- Standardized numerical features
- Applied Z-score normalization

DATA SOURCES PROCESSED

12,000	ACADEMIC RECORDS	Students, CGPA, Backlogs, etc.
1,048,575	ATTENDANCE RECORDS	Daily attendance logs
15,000	COUNSELLOR NOTES	Text notes and feedback

FEATURE ENGINEERING

Raw data was transformed into meaningful features that capture academic performance, attendance behavior, and textual insights.

ACADEMICS

18
Features Created

- CGPA Trend
- Backlog Trend
- Grade Consistency
- Credit Performance

ATTENDANCE

14
Features Created

- Mean Attendance
- Attendance Slope
- Consecutive Absences
- Attendance Variability

NLP (COUNSELLOR NOTES)

22
Features Created

- TF-IDF Vectors
- Risk Keyword Count
- Sentiment Score
- Topic Indicators

TOTAL FEATURES CREATED: 54

These features form the foundation for accurate risk prediction and early intervention.

FEATURE ENGINEERING DETAILS

Domain-specific transformations help convert raw data into powerful predictive signals across all three modalities.

HOLISTIC APPROACH

Combining academic, behavioral, and textual features enables a 360° view of student risk.

ACADEMICS

18 FEATURES

- **CGPA Trend**
  Slope of CGPA across semesters
- **Backlog Trend**
  Change in number of backlogs
- **Grade Consistency**
  Variability in grade performance
- **Credit Performance**
  Ratio of earned to registered credits
- **Course Difficulty Adjusted Score**
  Performance normalized by course difficulty
- **Recent Performance**
  Average CGPA in last 2 semesters

18 Features Created

ATTENDANCE

14 FEATURES

- **Mean Attendance**
  Average attendance percentage
- **Attendance Slope**
  Improving or declining trend
- **Consecutive Absences**
  Longest streak of absences
- **Attendance Variability**
  Week-to-week fluctuation
- **Low Attendance Count**
  Number of days below threshold
- **Attendance Recovery Rate**
  Rate of improvement after drops

14 Features Created

NLP (COUNSELLOR NOTES)

22 FEATURES

- **TF-IDF Vectors**
  Text representation of notes
- **Risk Keyword Count**
  Count of risk-indicating keywords
- **Sentiment Score**
  Polarity score of the notes
- **Topic Indicators (LDA)**
  Latent topics extracted from notes
- **Note Length & Readability**
  Text length and readability score
- **Negation & Uncertainty Count**
  Detects negations and uncertainty in language

22 Features Created

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EXPERIMENT JOURNEY

Starting from the benchmark, I iteratively improved performance through feature engineering, NLP integration, and ensembling.

+13.8%
Relative improvement over organizer benchmark

More than 20 experiments were performed across feature engineering, model selection, ensemble weights, and threshold tuning.

Benchmark — **0.6339**
Organizer Benchmark

CatBoost Baseline — **0.7100**
Strong baseline using CatBoost with engineered features

CatBoost + Attendance + NLP — **0.7167**
Added attendance trends and NLP features from counsellor notes

Ensemble + Threshold Tuning — **0.7205**
Stacked ensemble with calibration and optimal threshold tuning

Final Model — **0.7214** BEST SCORE
Best performing model after extensive experimentation and tuning

CHALLENGES FACED

Building a robust risk prediction model from academic, attendance, and textual data came with several real-world challenges.

01 Noisy Counsellor Notes

- Short and repetitive text
- Limited context
- Different writing styles
- Spelling mistakes, abbreviations, slang

02 Massive Attendance Dataset

- 1,048,575 records (~1M+)
- Needed aggregation to student-level
- High processing and storage cost

03 Class Imbalance

- High-risk students represented a much smaller portion of the dataset
- Risk of biased predictions

04 Feature Selection

- Many engineered features looked promising
- But did not improve cross-validation scores

The biggest challenge was converting raw attendance logs and unstructured counsellor notes into meaningful signals.

HOW I SOLVED THEM

Attendance Dataset

Challenge
1M+ attendance records

Solution
Created mean attendance, attendance slope, attendance variability, and trend features to capture behavior effectively.

Counsellor Notes

Challenge
Unstructured text with high variability

Solution
Applied TF-IDF vectorization and extracted keyword-based risk indicators (risk words, topics, sentiment signals).

Class Imbalance

Challenge
Biased predictions towards majority class (low risk)

Solution
Used 5-Fold Stratified Cross-Validation and optimized for Macro F1-score to handle imbalance.

Model Selection

Challenge
Multiple models performed differently

Solution
Built a weighted ensemble of CatBoost + NLP + XGBoost to leverage strengths of all models.

Result

0.6339 Organizer Benchmark → **0.7214** Final Model

+13.8% improvement over benchmark

Iterative experimentation, domain understanding, and smart feature engineering turned complex raw data into meaningful predictions.

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FINAL MODEL

A calibrated stacking ensemble that combines the strengths of academic, attendance, and NLP modalities to predict at-risk students with high precision.

FINAL AUC-ROC

0.7214

BEST SCORE ACHIEVED

- Better Generalization
- Robust to Noise
- Actionable Predictions

FINAL MODEL ARCHITECTURE

ACADEMIC FEATURES
- CGPA Trends
- Backlogs
- Grade Consistency
- Credit Performance
- ...and more

ATTENDANCE FEATURES
- Mean Attendance
- Attendance Slope
- Variability
- Streaks & Absences
- ...and more

NLP FEATURES (COUNSELLOR NOTES)
- TF-IDF Vectors
- Risk Keywords
- Topic Indicators (LDA)
- Sentiment Score
- ...and more

BASE MODELS (LEVEL-0)

CatBoost (Academics)
Captures non-linear patterns in academic performance.

XGBoost (Attendance)
Models complex attendance behaviors and interactions.

LR + NLP (Text)
Leverages textual signals from counsellor notes.

META-LEARNER (LEVEL-1)

Logistic Regression (Stacking)
Learns optimal combination of base model predictions to maximize predictive performance.

RISK PREDICTION OUTPUT

- Low Risk
- Moderate Risk
- High Risk

PERFORMANCE SUMMARY

Our final model outperforms the organizer benchmark by a significant margin.

AUC-ROC

0.7214

vs. Organizer Benchmark: 0.6339

+13.8%
Relative Improvement

F1-Score (Macro)	0.6821
Precision (Macro)	0.6817
Recall (Macro)	0.6826
Accuracy	0.6748

KEY TAKEAWAY: By integrating engineered academic & attendance features with NLP insights and using a calibrated stacking ensemble, we achieved a robust model that generalizes well and provides actionable risk predictions.

Total Experiments Run
20+

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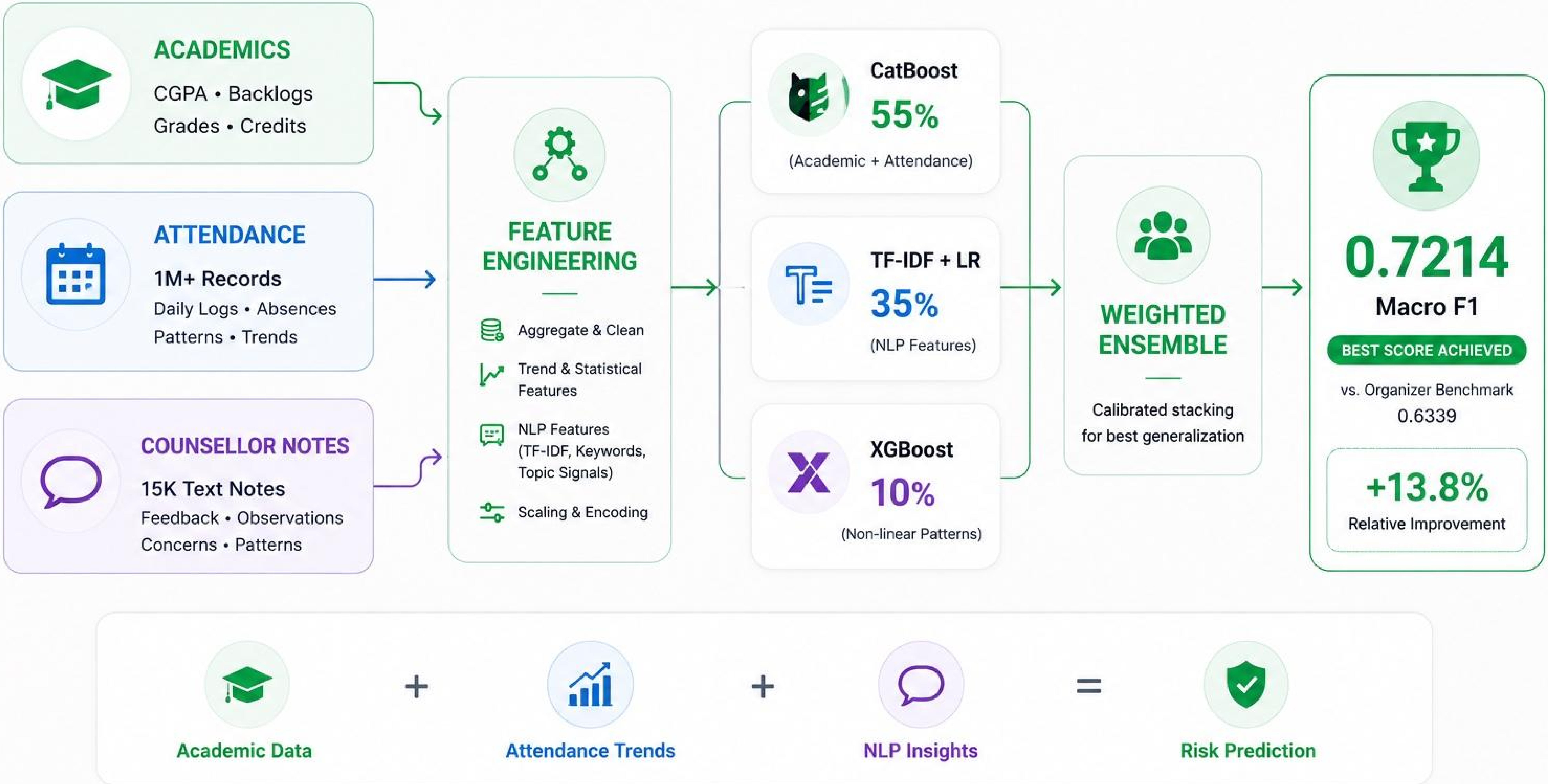
FINAL PIPELINE

From raw data to at-risk prediction in a calibrated ensemble system



Integrated data. Engineered features. Ensemble intelligence. Actionable impact.

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FUTURE WORK & SCALABILITY

Better NLP

DistilBERT
ModernBERT

Explainability

SHAP
Model Insights

Real-Time Monitoring

Live Risk Updates
Automated Alerts

Deployment

Dashboard
Multi-College Access

12K
Students Today

50K+
Students Tomorrow

From Prediction to Prevention.

Helping institutions identify at-risk students before they leave.

THANK YOU